450MHz Alliance Overview

2022

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promoting the advancement of 450 MHz worldwide.

Overview of 450MHz Alliance

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The 450MHz Alliance:

» Industry association representing the interests of 450MHz spectrum stakeholders

» Members: wireless operators, equipment and device manufacturers, companies representing various vertical markets for machine-to-machine communication

» Goal: sharing best practices and stimulating the 450MHz ecosystem globally

» Improve the usage and attractiveness of the 450MHz band worldwide

Objectives of 450MHz Alliance

» Ensure suitable *regulatory measures* (i.e. FCC, CEPT, 3GPP)

» Advertise successful *business models* as templates for other operators, countries and regions

» Ensure availability of *emerging technologies* (i.e. LTE 450/410, LTE-M, Band 31, 72, 87 and 88 extension to 380-400MHz)

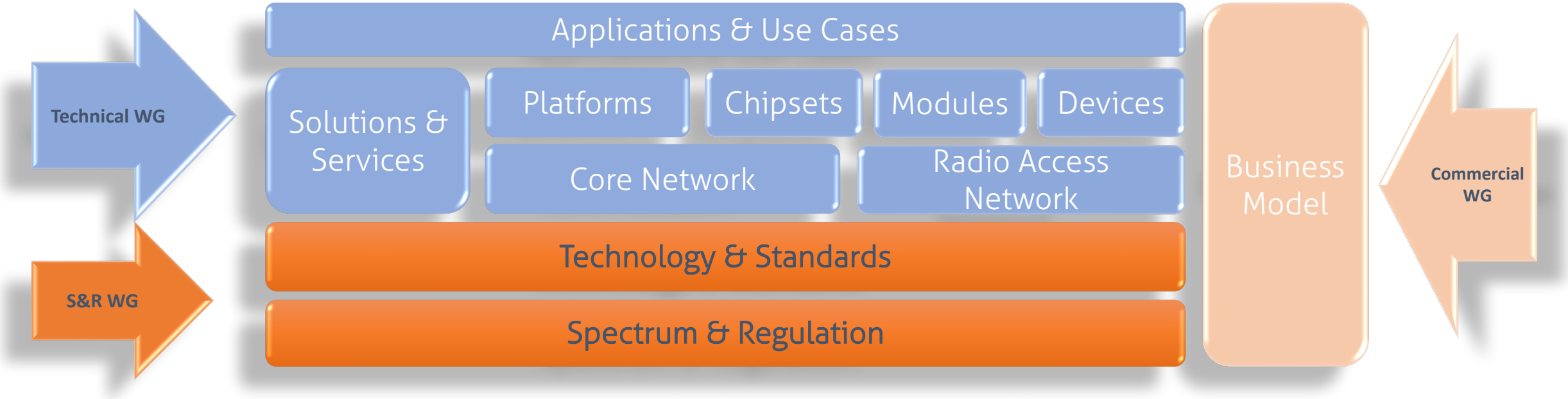
» Ensure increased *availability of ecosystem* (infrastructure, chipsets, modules, end devices, etc.) for a wide range of applications (e.g. IoT, PPDR, MBB)

» Share *technical knowledge* and *best practices*

450 MHz Alliance - Work structure

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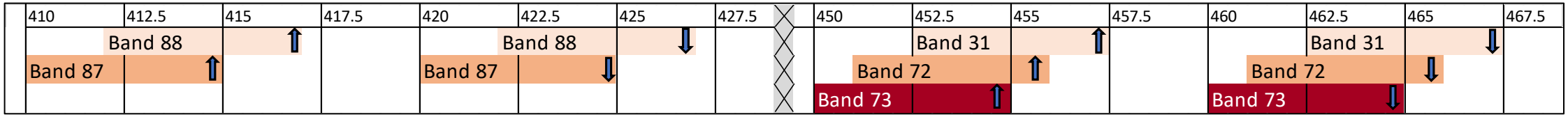
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3GPP Band plan

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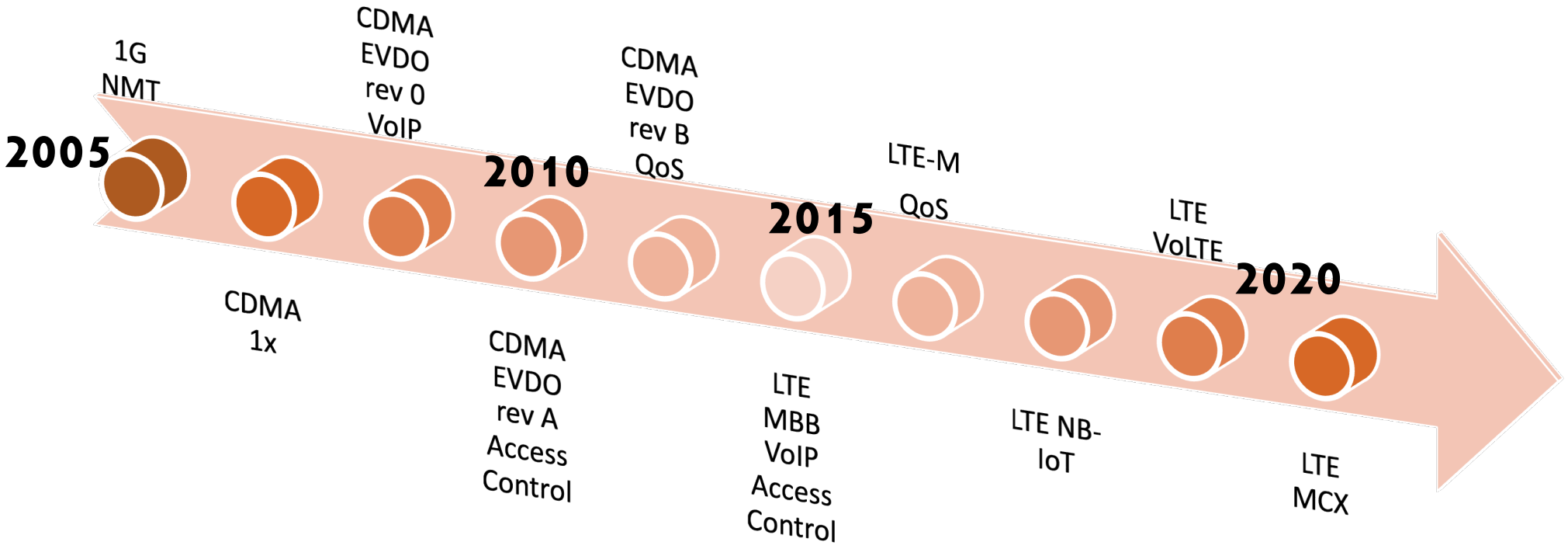


- Two bands in 410 MHz, one high and one low
- Two bands in 450 MHz, one high and one low
- Band 73 not in use
- 380 MHz work in progress

Evolution of Services in the 400MHz

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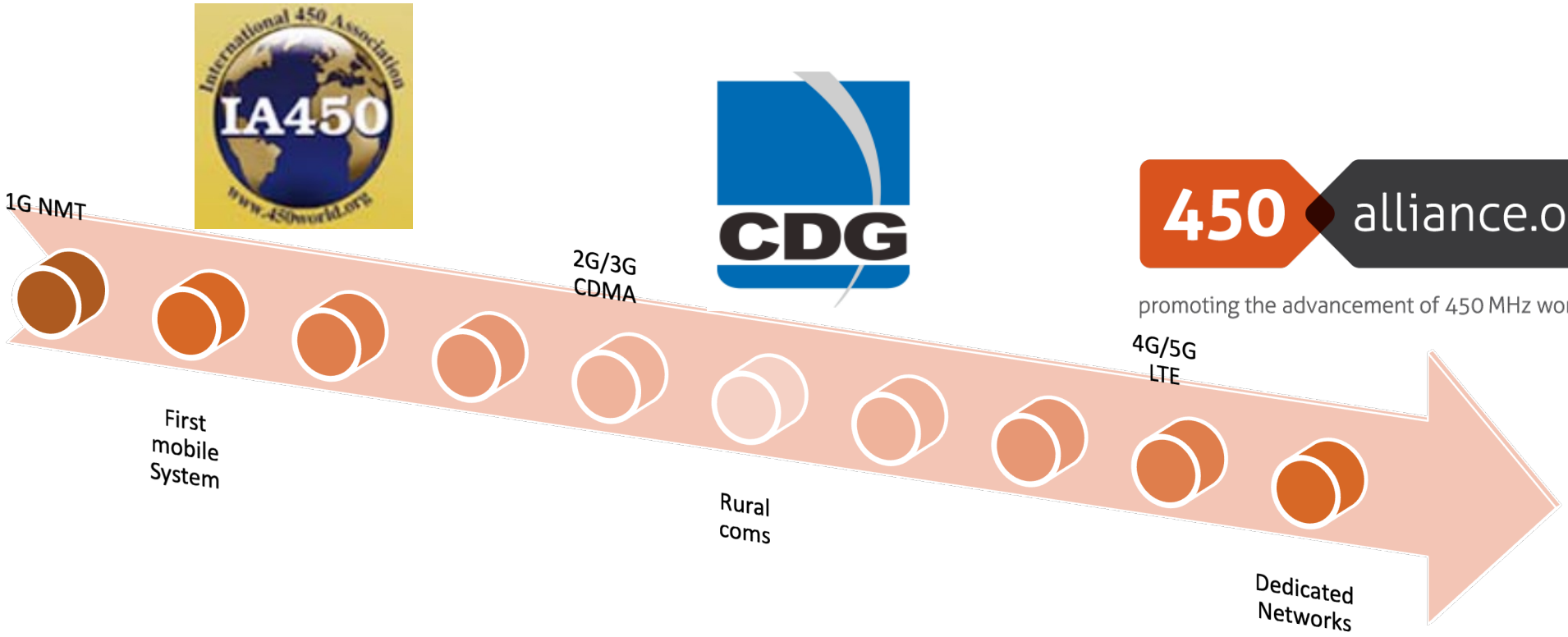
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History

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Our Members

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Our Partners

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Success Factors for 450MHz Industry

- 450MHz spectrum benefits & global availability to drive standardization & ecosystem
- Standardized mobile technology and attractive device ecosystem to be competitive with mass market mobile operators in key parameters
- Differentiated niche business model with strong unique selling propositions
  - (i) outside the capacity play given limited spectrum
  - (ii) but with large enough device volume from IoT applications to fuel ecosystem

Now All Coming Together!

Overview of 450MHz Networks Worldwide

Markets Worldwide as of Q3 2021

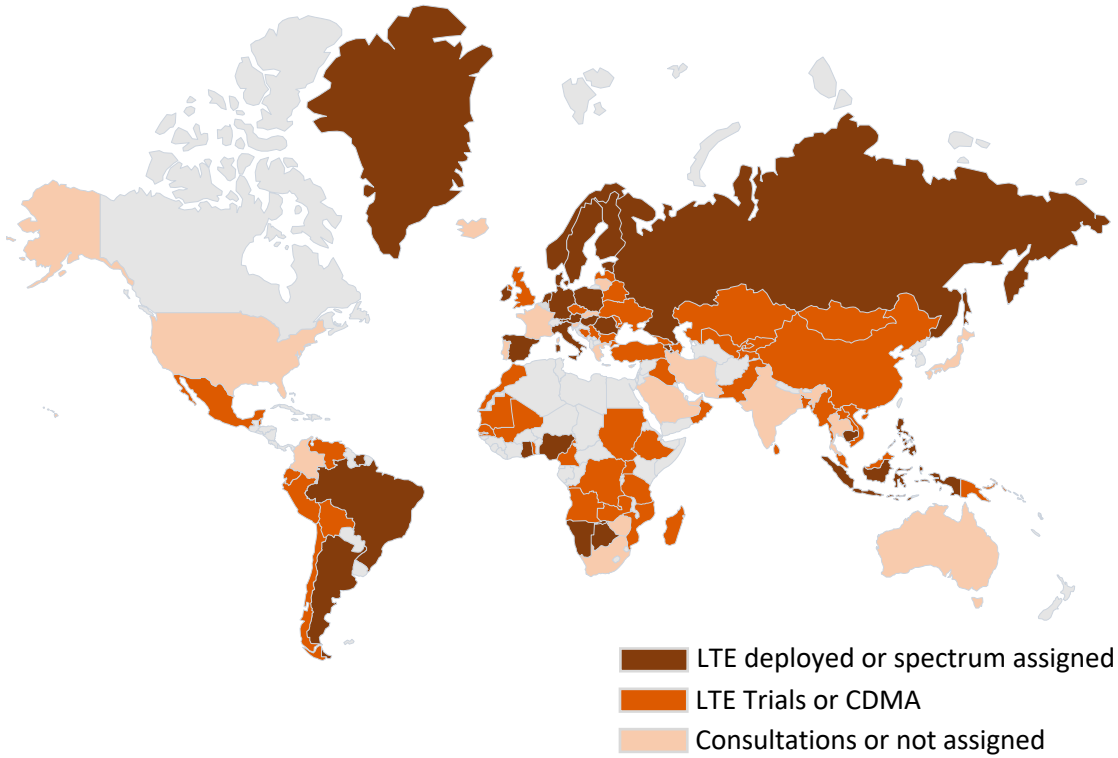
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» 111 Countries confirmed status

» 82 Regulators responded

	Deployments	LTE Deployments	Consultation	Not in use
Africa	21	4	1	1
Americas	9	3	2	1
Asia and Oceania	18	2	5	5
Europe	26	15	4	9
Total	74	24	12	16



Spectrum available or in consultation includes **6.6 billion** people or **85 percent** of the global population
Network available or in trial includes **4.1 billion** people or **52 percent** of the global population

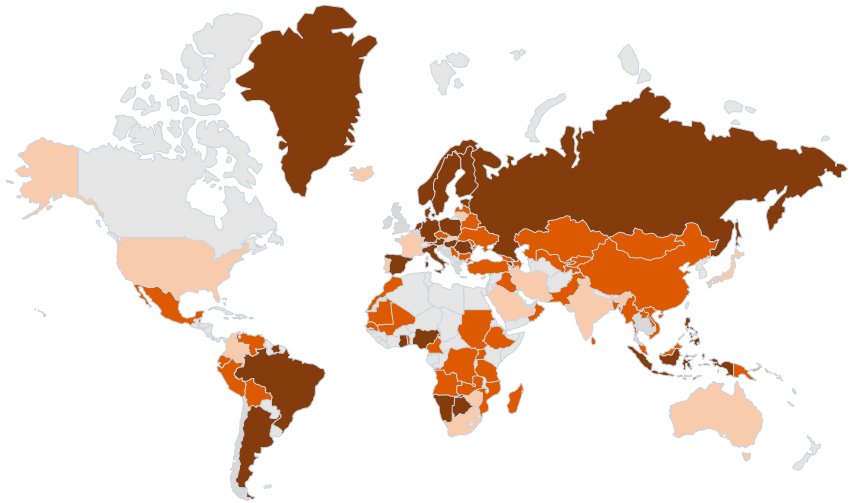
The above map is subject to change

Markets Worldwide as of Q3 2021

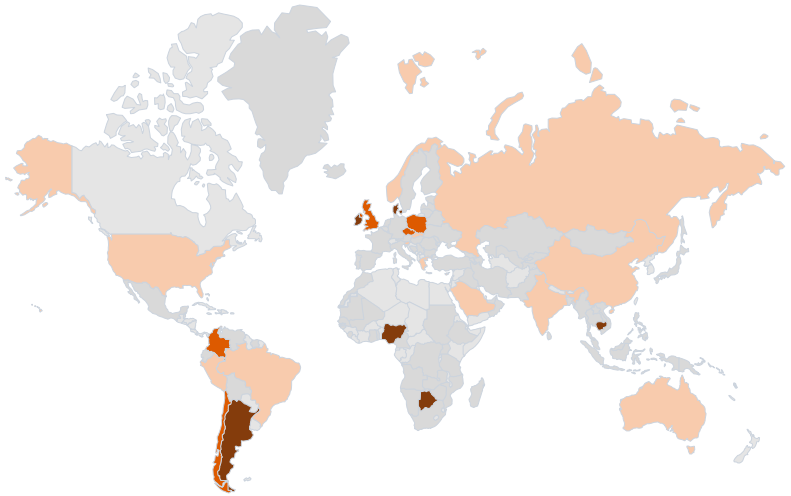
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450 MHz

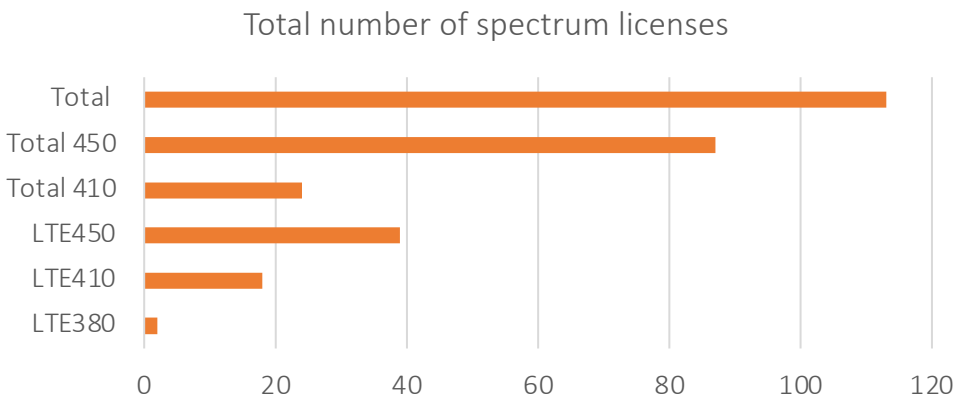


410 MHz



380 MHz





Spectrum available or in consultation includes **6.6 billion** people or **85 percent** of the global population

Network available or in trial includes **4.1 billion** people or **52 percent** of the global population

LTE deployed or spectrum assigned

LTE Trials or CDMA

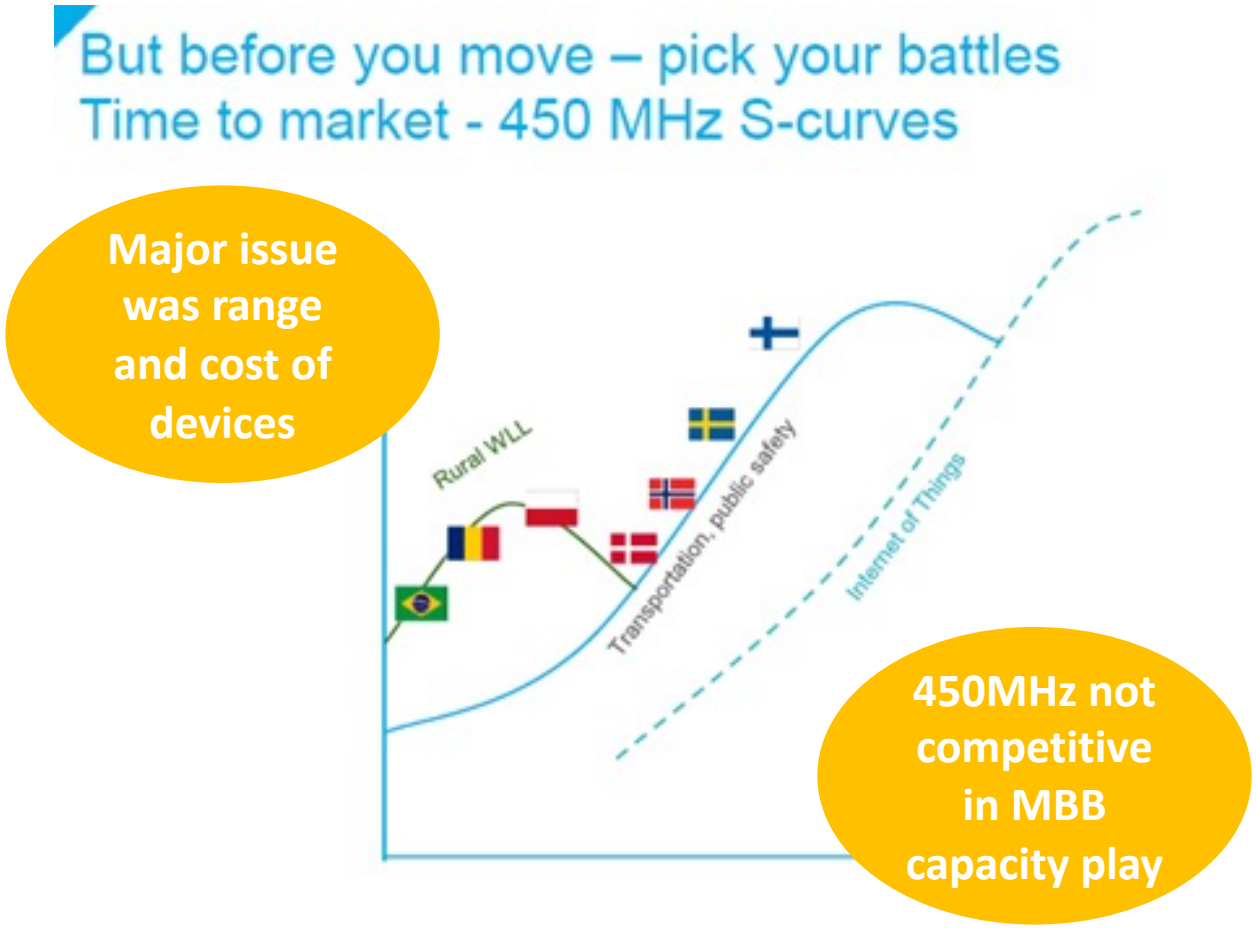
Consultations or not assigned

The above maps are subject to change

450MHz Ecosystem, Business Model, Case Studies & Market Potential

Evolution of 450MHz Business Model

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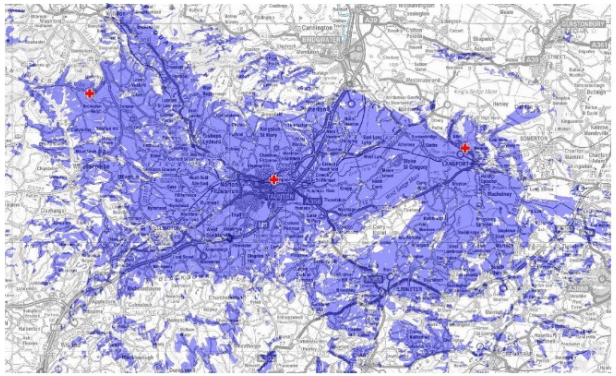
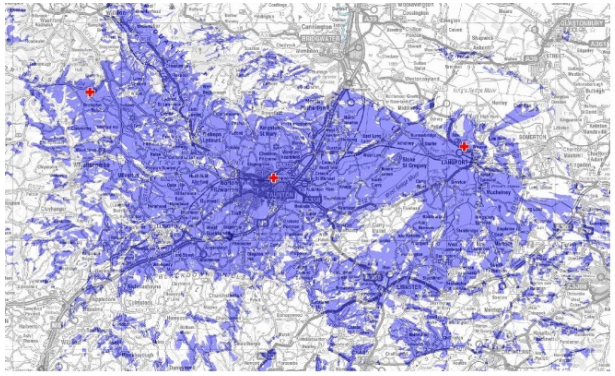
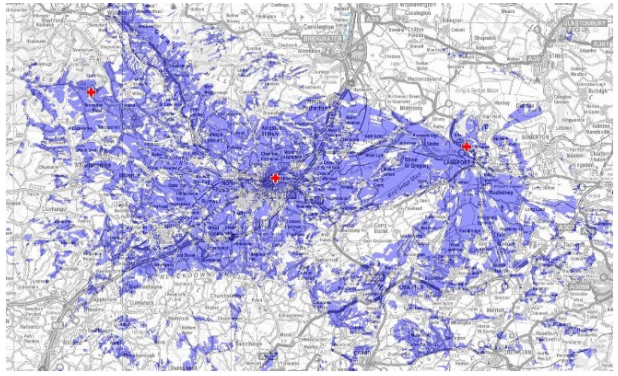
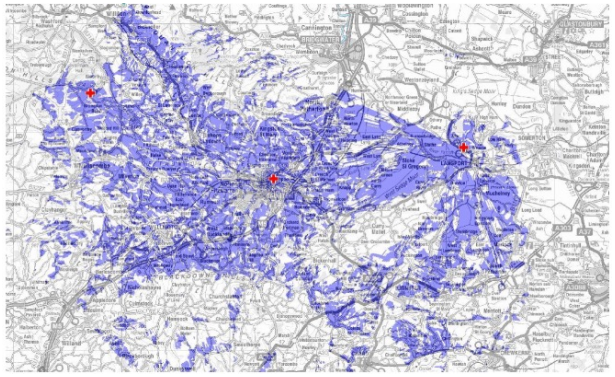
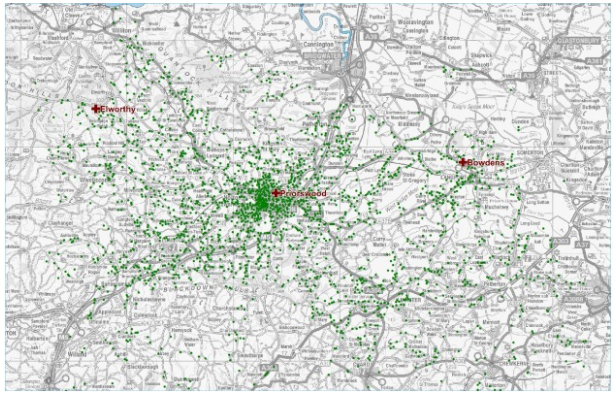
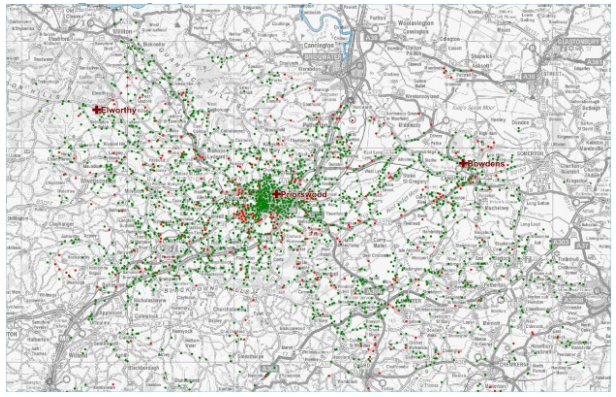
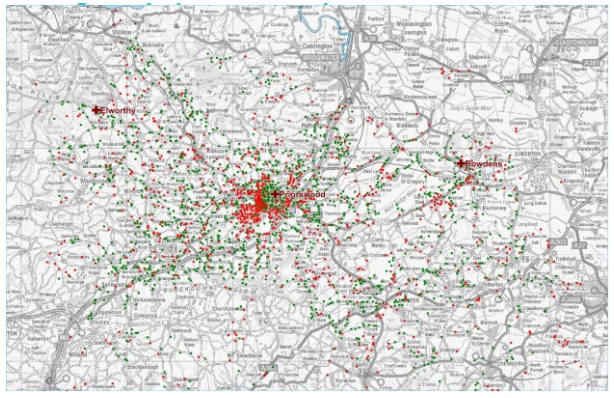
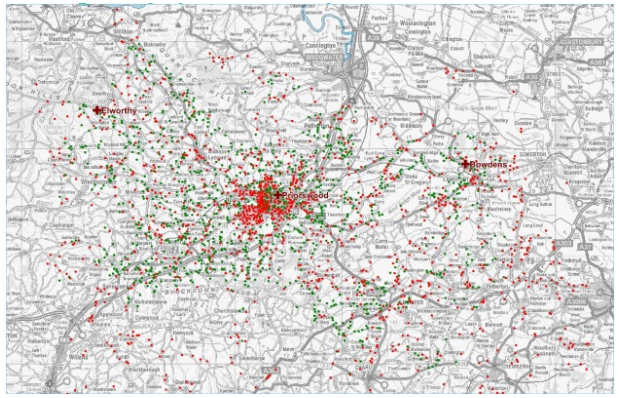
- Originally first public mobile networks in 450MHz in the early 1990's (NMT450)
- From 2001 deployments of CDMA450 chiefly for low cost rural coverage in countries with low population coverage / developing world
- Increasing focus on rural broadband services to address digital divide in rural areas/emerging markets
- Use for dedicated services such as public safety & security in the developing world
- Global trend for refarming of 450MHz spectrum to address new LTE use cases like critical M2M/IoT and emergency voice and data communication for companies and organizations (shared PMR/PAMR networks)

Source: Ukkoverkot

Coverage

WESTERN POWER DISTRIBUTION
Serving the Midlands, South West and Wales

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410 MHz	740 MHz	2350 MHz	3850 MHz
			
			
100% connected	91,3% connected 2.5-3 x eNB	58,7% connected 12-15 x eNB	45,6% connected 20-25 x eNB

450MHz's USPs for vertical markets

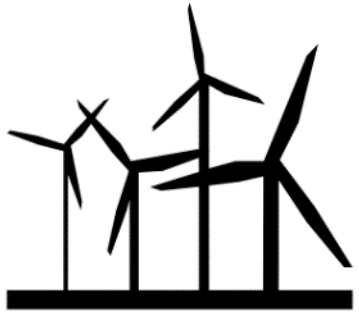
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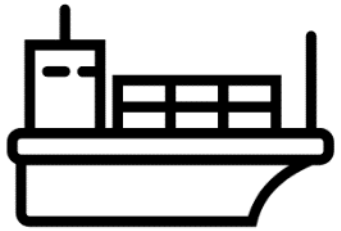
- Benefit of superb coverage with fewer sites translating into lower network cost and ability for fast deployment
- Ability to build and operate highly resilient and secure network fulfilling the requirements of business + mission critical applications at reasonable cost
- Availability of standardized and scalable LTE technology that is becoming the mainstream solution for IoT with eMTC and NB-IoT enhancements
- Only frequency band supporting >95% coverage probability for time critical M2M services deep indoor

➢ **Much better service proposition for critical voice / M2M applications than large public mobile networks (and outside the capacity game)**

➢ **At competitive cost due to low network cost and comparable device cost**



Utilities



Logistics



Emergency call



Agriculture

LTE450 supports both critical and less critical M2M / IoT applications

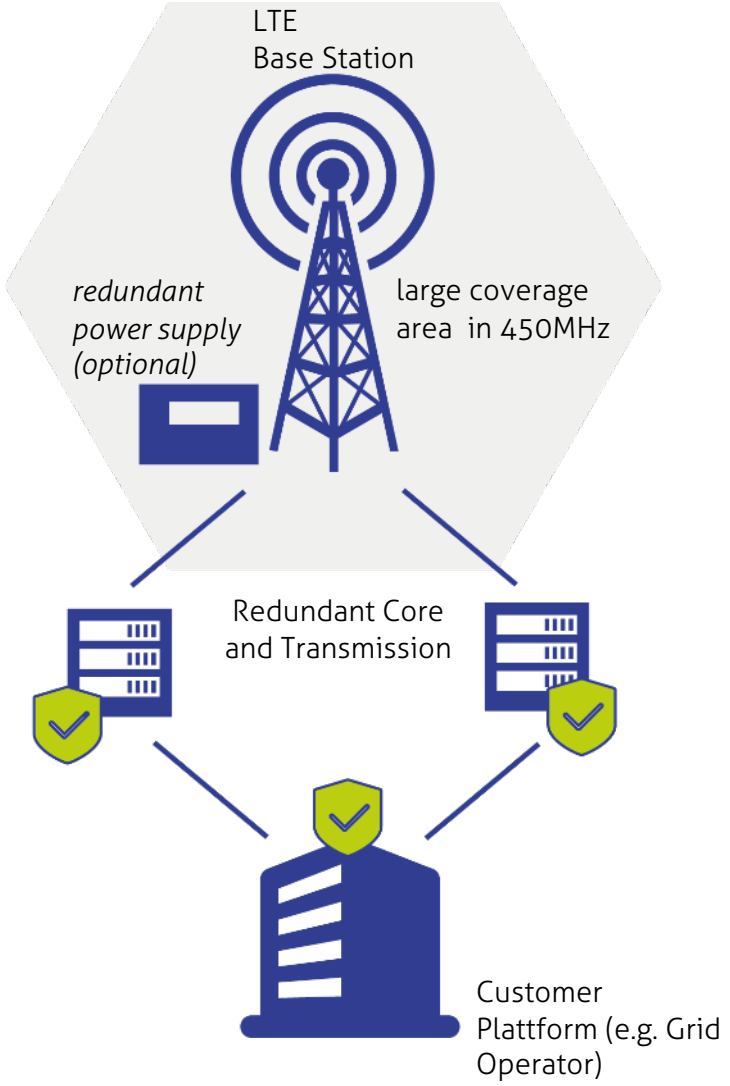
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LTE450 can be optimized for **M2M** communications and closed user groups

Standardised and **proven** mobile communications technologies to achieve high **performance** (capacity, latency, priority handling, scalability)

Control on lifecycle management by long term commitment on technology standards

Most connected devices will be in the field for 10-15 years



Large **coverage** area and deep **indoor penetration**

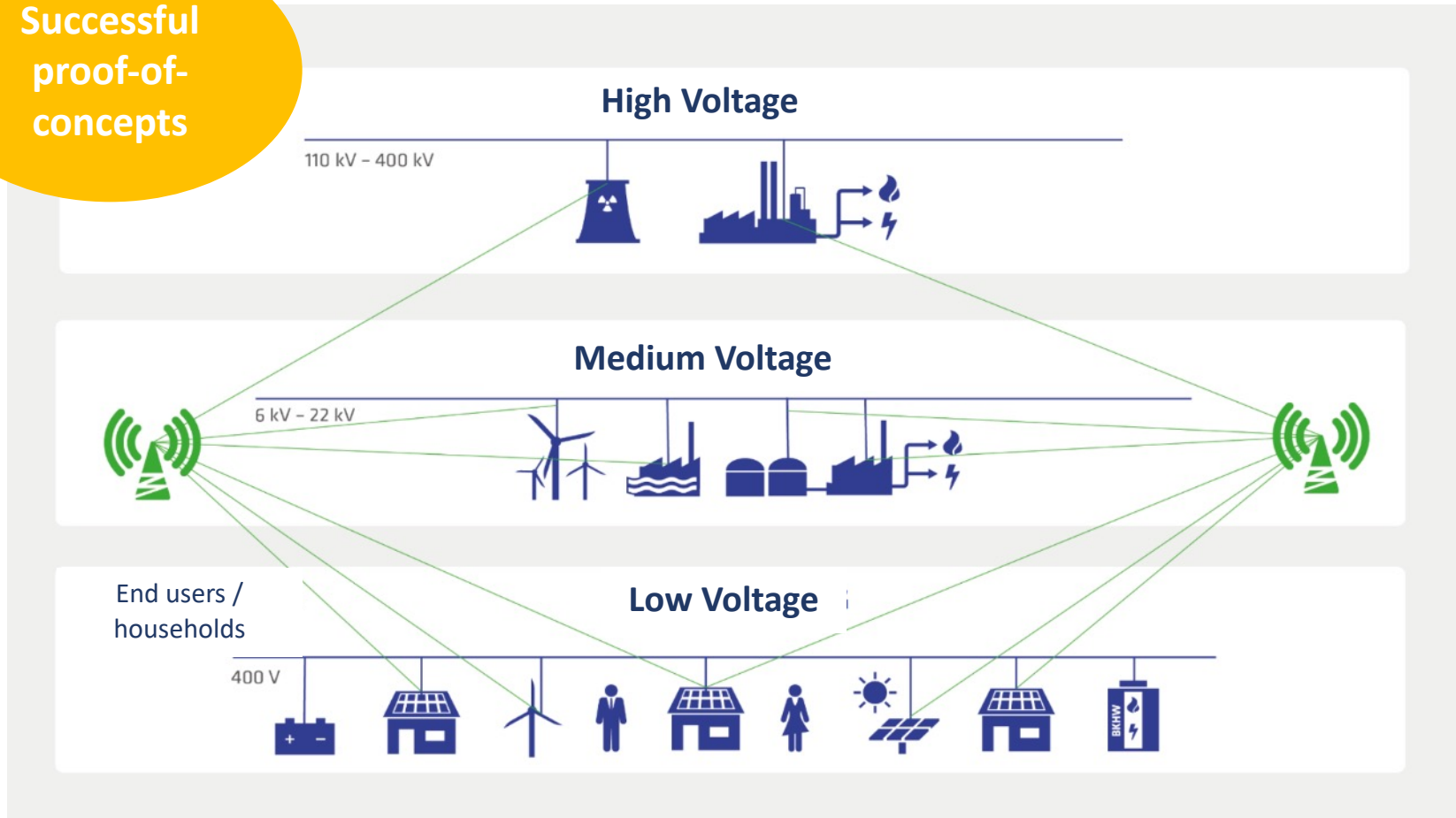
Power supply backed up using batteries and generators to continue services during (major) power outages

Stand alone network without public mass market services, allowing for high **resilience** and **security**

Utilities as driving force for 450MHz

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Successful proof-of-concepts



Application of 450MHz connectivity for utilities

- Smart Grid for Electricity and Gas networks
- Smart Meters and/or Smart Meter Gateways
- Emergency voice and data communications
- Others, such as EV charging stations, prosumers, etc.
- Upside for further applications in district heat & water supply networks

Further IoT/M2M use cases – smart city



450MHz networks are also great fit for the various M2M/IoT applications due to the **enhanced indoor coverage** compared to higher bands

Thanks to the availability of the worldwide ecosystem, 450MHz modules can be **easily integrated into new applications** e.g. in the area of smart city

Primary applications for the 450MHz lie in the street lightning applications, control and monitoring of diverse environmental sensors, traffic control and so forth.

Smart city projects are one of the business areas of the Austrian 450MHz network operator – ArgoNET but also Dutch Utility Connect

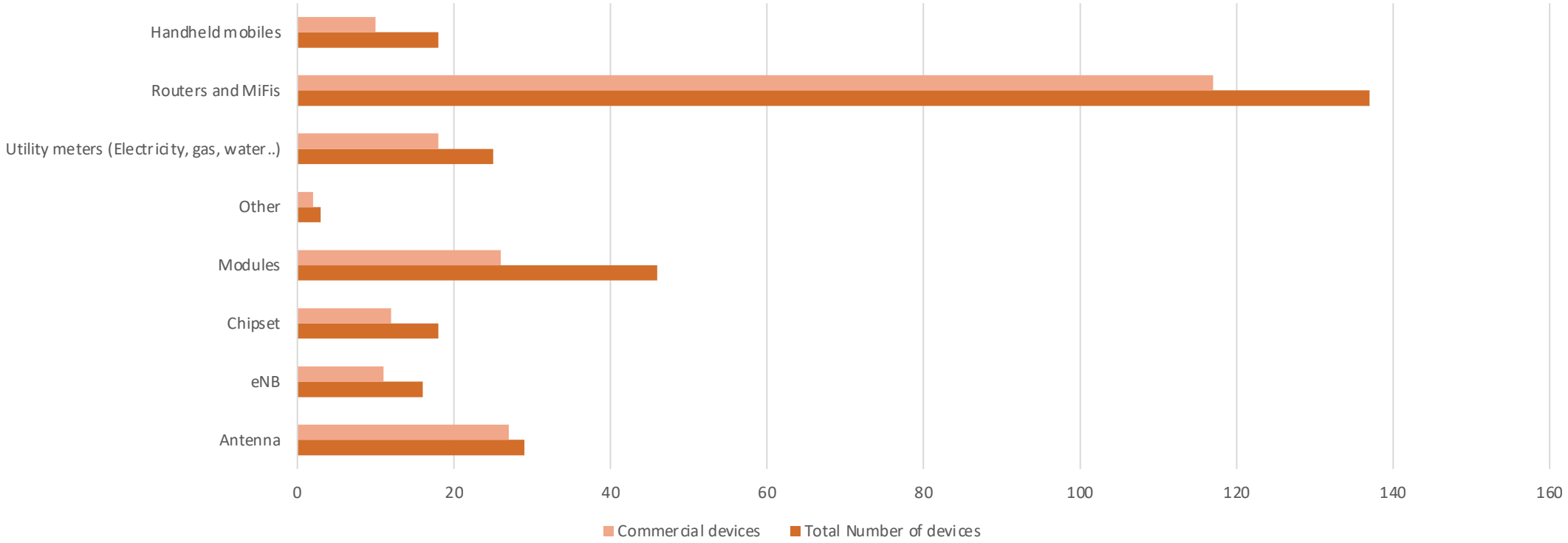
Cities

- Traffic Management
- Environmental Monitoring
- Grid applications

Device update 2021

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» Device total of 183 where of 147 commercially available



» The listed devices are verified with the suppliers

» Most devices multiband and multi technology

Growing LTE450 Ecosystem

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- Networks





ERICSSON, NOKIA, Hytera, AIRBUS, CableFree Wireless Excellence, HUAWEI, ZTE中兴, Parallel WIRELESS, Anktion

- UE Chipsets



QUALCOMM, altair A SONY Company, SEQUANS COMMUNICATIONS, UNISOC, GCT, ublox, ASR MICROELECTRONICS, HISILICON, SANECHIPS, MEDIATEK

- M2M Modules





AsiaTelco TECHNOLOGIES, Telit, HUAWEI, GOSUNCN, UNITAC, NTmore Network Technology, Cellient Smart & Green, QUECTEL Wireless Module Expert, NOTION, SIMCom a SUNSEA AIOT company, SMAWAVE, THALES, ublox, SIERRA WIRELESS, neoway, Fibocom

- LTE450 ecosystem is maturing & further growing taking advantage of global LTE developments (e.g. LTE-M, NB-IoT, MC-PTT)
- Market is further growing due to spectrum refarming activities
- 450MHz ideally suited for M2M / IoT and critical voice applications
- There are a mature range of equipment for the networks
- Chipsets and Modules support for eMTC & NB-IoT with multiband capabilities from 410-2600MHz are commercially available

The above table is subject to change

Growing LTE450 Ecosystem

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- Handhelds



HUAWEI, NOKIA, AORO, Hytera, SMAWAVE, RugGear, Caltta, UNITAC, KZ TECH

- Routers



intelliport, Anktion, GARDEROS, NOKIA, ANDRA, VIRTUAL ACCESS, Caltta, CISCO, DIGI, UNITAC, SMAWAVE, encore networks, Cellient, NOTION, simoco wireless solutions, ADVANTECH, HUAWEI, AsiaTelc, NTmore

- Meter and Other



NOKIA, intelliport, PPC Power Plus Communications, WM SYSTEMS, SANXING, KAIFA, ANDRA, xemex

- The device range is growing in line with global operations
- Handhelds are mainly supporting an industrial and rugged design
- Availability of routers are good and are growing with the implementation of modules within new router and gateway families
- eMTC and NB-IoT have extended the range of products to include meters for electricity, gas and water and other sensor and gateways with requirements for low power consumption

The above table is subject to change

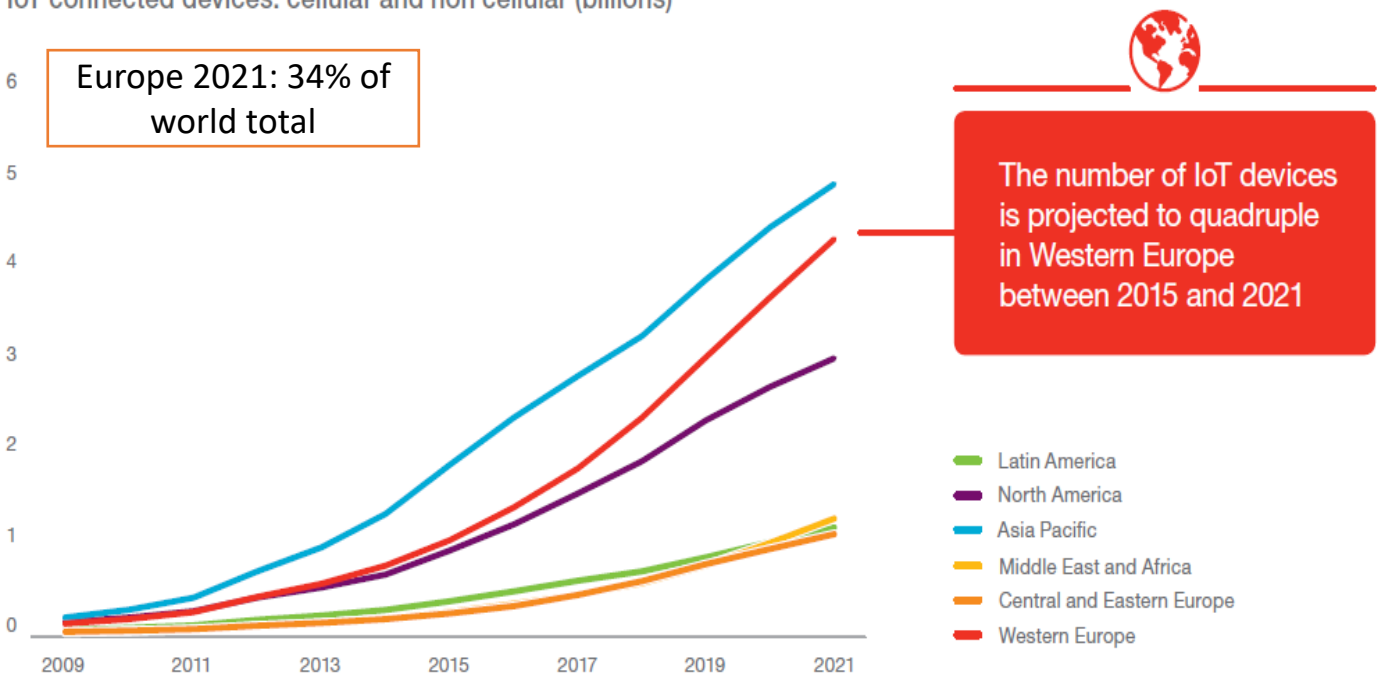
World market for cellular IoT will grow to 1,5 billion devices, of which 34% in Europe

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The Internet of Things (IoT) is expected to surpass mobile phones as the largest category of connected devices in 2018

In 2019, LTE will be the dominant mobile access technology

Source: Ericsson Mobility Report 2016



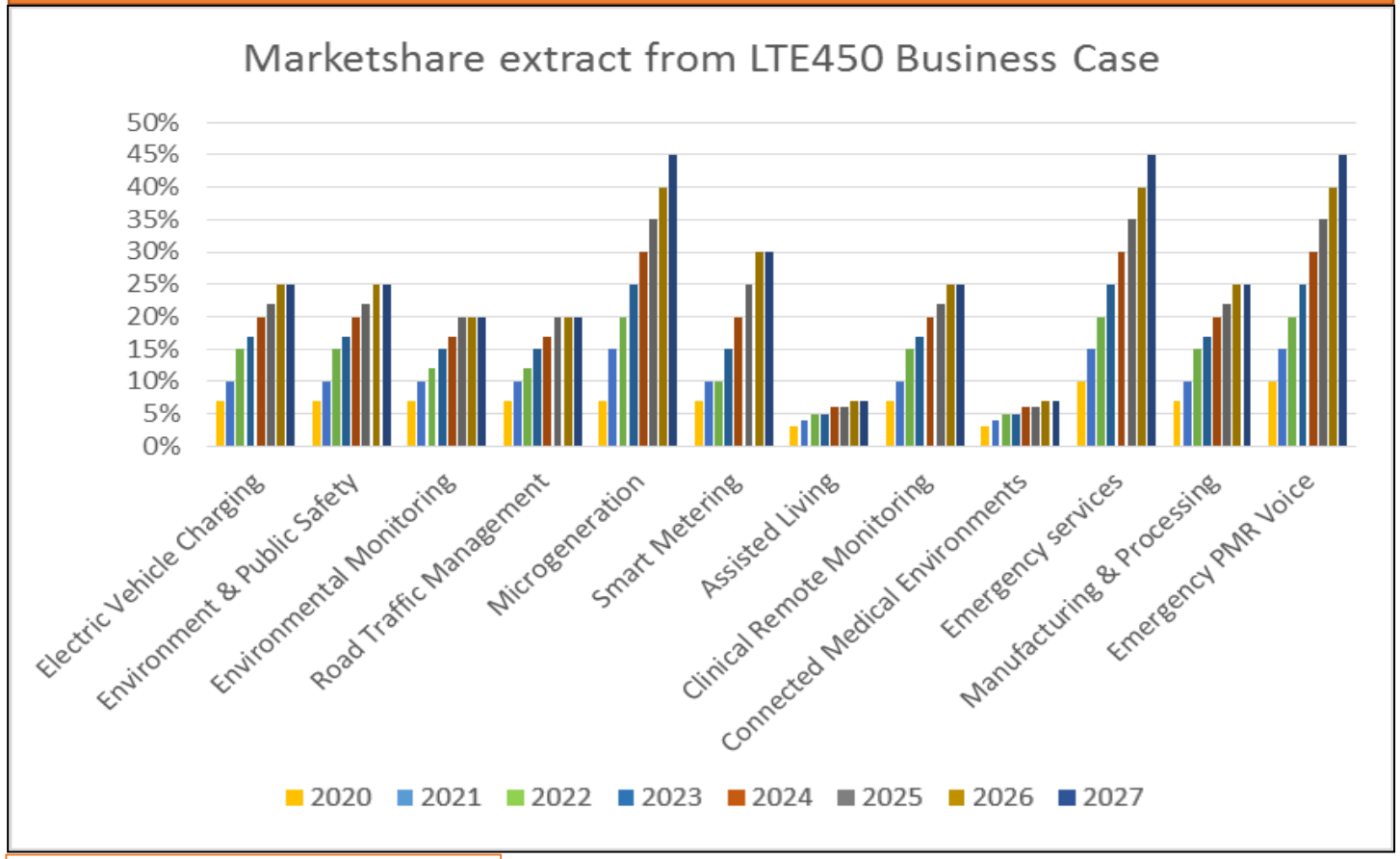
	15 billion	28 billion	CAGR 2015–2021
Cellular IoT	0.4	1.5	27%
Non-cellular IoT	4.2	14.2	22%
PC/laptop/tablet	1.7	1.8	1%
Mobile phones	7.1	8.6	3%
Fixed phones	1.3	1.4	0%
	2015	2021	

- TAM 2021: 1.5 billion connections worldwide for cellular IoT
- eMTC and NB-IoT will allow to address substantial share of LoRA type networks (non-cellular IoT)

Based on market share forecasts, the gross adds of LTE450 IoT devices 2020-2024 will be 32M

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Sample forecasted market share development for a variety of IoT applications



Source: European 450MHz Operator

*calculation of population living in the countries with 450MHz license granted for PMR/IoT solutions ("IoT focus") and countries with 450MHz for PPDR ("IoT potential"), compared to whole European population

Estimate of Total Addressable Market Europe through 2024	Countries with LTE450 IoT focus	Countries with LTE450 IoT potential	Total Europe
Gross adds IoT Cellular & LPWA through 2024	1.005M		
Share of population with LTE450 IoT focus (27%) respectively potential (31%)	**27%***	**31%***	**58%**
Addressable Market	**270M**	**310M**	**580M**
Business critical & smart meters (37%)	100M	115M	215M
Average market share for LTE450 networks 2020-2023	20%	10%	14,7%
Forecasted gross adds LTE450 2020 – 2024	**20M**	**12M**	**32M**

The figure of 20M respectively 32M does NOT include:

- The rest of the world (especially China, Brazil, Japan, Indonesia, Philippines, Asian population of Russia and Turkey)
- Public Safety Networks (PPDR)
- Military Applications
- Mobile Broadband

See next slide for an estimate of the global market potential

450MHz Alliance Study March 2019
Worldwide market potential until 2030

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2020 2024 2030

Narrow Potential Europe:
European countries with IoT focus, only IoT Potential

- TAM IoT until 2024: 270M
  - Business Critical 37%: 100M
- IoT Market Share 20%
- **LTE 450 Devices: 20M**

- TAM IoT until 2030: 4.500M
  - Business Critical 20%: 900M
- IoT Market Share 25%
- **LTE 450 Devices: 225M**

- In typical 4-operator market LTE450 offers better product given security, resilience and coverage, i.e. 25% market share well justified

Expanded Potential Europe:
All European countries with 450MHz licenses, including also PPDR, MBB & Military potential

- TAM IoT until 2024: 580M
  - Business Critical 37%: 215M
- IoT Market Share 14,7%: 32M
- MBB, PPDR, Military: 15M
- **LTE 450 Devices: 47M**

- TAM IoT until 2030: 9.600M
  - Business Critical 20%: 1.920M
- IoT Market Share 15%: 288M
- MBB, PPDR, Military: 30M
- **LTE 450 Devices: 318M**

Extrapolated Global Potential:
108 countries expected with LTE 450 MHz licence, potential including IoT, PMR, PPDR, MBB, Military

- TAM IoT until 2024: 1.630M
  - Business Critical 37%: 603M
- IoT Market Share 5%: 30M
- MBB, PPDR, Military: 45M
- **LTE 450 Devices: 75M**

- TAM IoT until 2030: 30B
  - Business Critical 20%: 6.000M
- IoT Market Share 7%: 420M
- MBB, PPDR, Military: 90M
- **LTE 450 Devices: 510M**

Closing remarks

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- Global and increasing footprint of 450MHz networks and new spectrum licenses fostered by availability of LTE technologies for 450MHz
- Strong business case for critical IoT communication leveraging off superior propagation and related cost benefits for dedicated, highly secure and resilient networks
- Initial focus of 450MHz in Europe driven largely by utility requirements with large potential in other critical sectors such as health, transport, agriculture and public services with global potential as secure & reliable communication essential for world-wide digitization of critical processes and infrastructure
- High availability and security in dedicated 450MHz networks enabling value for money positioning against public cellular networks resulting in higher accepted TCO per device allowing for reasonable device/module premium
- LTE450 for IoT market expected to start in 2020 with strong growth from 2021 onwards
- LTE band 72 developing in key markets on top of existing band 31 networks

⇒ 450 MHz benefits from globally standardized technology with growing attractive device ecosystem

⇒ Differentiated niche business model with strong unique selling propositions

Thank you!

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